Thermo-Credit Theory: A Credit-First Thermodynamic Mapping

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Abstract

This note develops Thermo Credit as a measurement-first macro-credit model of how the destination of new credit alters the transformation of credit expansion into output, inflation, asset-price acceleration, and fragility. The framework combines (i) an SFC-style conservation backbone based on balance-sheet identities with (ii) a non-equilibrium "credit thermodynamics" layer in which irreversibility and hysteresis are first-class objects. We introduce a minimal state description built around credit partition, capacity, and stress, including an internal bookkeeping potential $U(S_{\mathrm{M}}, V_{\mathrm{C}}, \xi, \ldots)$ with memory variable ξ, and we define operational gauges such as Helmholtz-style free energy F_{C} and exergy-like availability X_{C}. Maxwell-like integrability is treated as a local-equilibrium limit rather than a law; persistent violations are interpreted as measurable non-equilibrium "vorticity" in the $(S_{\mathrm{M}}, V_{\mathrm{C}})$ state plane. The construction is deliberately practice-first: quantities are intended to be computable from public data, judged by forecast usefulness and falsifiability, and distributed in recalculable form through specification, code, and AI-facing interfaces rather than through narrative alone.

1 Problem Statement and Scope

Classical money-first views such as the Quantity Theory of Money (QTM) explain nominal dynamics in terms of a money stock, its velocity, and shocks or policy actions [1, 2, 3]. More recent credit-first views, notably the Quantity Theory of Credit (QTC), emphasize that banks create deposits when they lend and that the use of credit (real vs. financial, productive vs. speculative) matters for macro-financial outcomes [4, 5]. In parallel, both information theory and statistical mechanics have inspired analogies for income and wealth distributions [6, 7, 8, 9, 10].

This note takes a minimal and operational step: we build a thermodynamic-style state description that (i) separates scale and dispersion of monetary/credit uses, (ii) introduces an explicit capacity variable for bank balance sheets, and (iii) yields testable constraints and early-warning gauges. The aim is not to claim that macro data obey physical laws, but to provide a disciplined bookkeeping analogy useful for supervision, stress testing[K18], and systematic monitoring.

Research target and use cases

The measurement-first revision narrows the framework to one research question: how does the destination of new credit alter the transformation of credit expansion into growth, inflation, asset-price acceleration, and fragility? Operationally, the current version is designed for three immediate use cases:

1. early-warning signals for asset inflation and systemic fragility;
2. description of the real-versus-asset credit mix rather than credit volume alone;

3. forecast improvement relative to total-credit-only baselines.

This keeps the theory tied to a concrete mapping from inputs to outputs and prevents the thermodynamic analogy from drifting into a purely descriptive vocabulary.

Related Work

Several strands of research motivate the thermodynamic-style mapping used in this note. First, high-frequency microstructure studies have explored temperature- and entropy-like summaries of order-book dynamics. For example, Li et al. (2024) define "market temperature" and "market entropy" from depth, imbalance, and turnover in limit-order books, showing that these quantities track short-term liquidity conditions and stress episodes. These approaches remain at the micro (tick-by-tick) level and do not involve credit allocation or regulatory constraints.

Second, Feng (2019) develops a thermodynamic representation of model risk using entropy flows and worst-case expectations. This work treats market uncertainty as an information-theoretic source of dissipation within pricing models. It is conceptually related through the shared use of temperature- and energy-like ideas, but it does not address credit capacity, bank balance sheets, or sectoral allocation.

The present note differs in scope: we apply a coarse-grained, macro-credit state description aimed at supervision and early-warning applications. Our variables (S_M, V_C, T_L, p_C) are built from balance-sheet aggregates and dispersion measures, not from high-frequency order-book dynamics or pricing-model uncertainty. Loop area in the (p_C, V_C) plane plays the role of dissipation at the macro-credit level, and the quality-factor construction introduced here has no direct analogue in existing microstructure or model-risk thermodynamic papers.

2 Credit Partition and State Variables

We work over a chosen system (jurisdiction, sector set) and time aggregation (e.g. monthly). Let M_{in} denote money-in-circulation[K10] over this domain, and let $\{q_i\}$ be a stable, MECE (Mutually Exclusive, Collectively Exhaustive)[K15] partition of its uses (real activity, housing credit, securities margin, etc.), with

$$\sum_i q_i = 1.$$

2.1 Measurement-first overlay: credit partition and targets

To make the framework operational, we add a minimal measurement-first overlay whose purpose is narrower than the full thermodynamic analogy. The immediate question is: how does the destination of new credit change the transformation of credit expansion into output, prices, asset prices, and fragility?

The smallest useful state vector is

$$\left(C_t, q_t, Y_t^N, Y_t^R, P_t, A_t, S_t\right),$$

where C_t denotes new credit flow during period t, $q_t \in [0,1]$ is the share of that flow directed to real activity, Y_t^N is a nominal output proxy used for accounting ratios, Y_t^R is a real-activity target, P_t is a broad price-level proxy, A_t is an asset-price or balance-sheet accumulation proxy, and S_t is a stress / fragility state. The coarse real-versus-asset split is then

$$C_t^R = q_t C_t, \qquad C_t^A = (1 - q_t) C_t. \tag{1}$$

This partition share q_t should not be conflated with the entropy shares $\{q_i\}$ used below. The scalar q_t classifies *new credit flows* into real and asset destinations, whereas $\{q_i\}$ partitions broader money/credit uses for the entropy construction. They are related empirical objects, but they are not identical.

Identification of q_t is best treated in layers: (i) direct observation from sectoral lending and loan-purpose data when available, (ii) proxy inference from housing, equity, investment, and stress variables when direct splits are incomplete, and (iii) latent-state estimation only after the reduced-form version is stable.

2.2 Monetary dispersion entropy

We define an entropy-like extensive index

$$S_{\mathrm{M}} = kM_{\mathrm{in}}H(q), \qquad H(q) \equiv -\sum_i q_i \log q_i, \tag{2}$$

where $k > 0$ is a conventional scaling constant. This mirrors ideal mixing entropy $\Delta S_{\mathrm{mix}} \propto NH(x)$ and follows the information-theoretic form of Shannon [9]. S_{M} grows with both scale and dispersion; concentration (reallocation into fewer uses) reduces $H(q)$ and can reduce S_{M} even at fixed M_{in}.

Formally, $H(q)$ is the Shannon–Boltzmann entropy of the MECE bucket probabilities $\{q_i\}$ with natural logarithm (no Tsallis, Rényi, or alternative families are used here). In addition to the extensive S_{M}, the implementation tracks a scale-free version $S_{\mathrm{M,hat}} = H(q)/\log K$ (with K buckets) and per-bucket contributions $S^{(i)}_{\mathrm{M,in}} = kM_{\mathrm{in}}\tilde{q}_i$. For the baseline JP configuration these buckets correspond to "productive", "housing", "consumption", "financial", and "government" allocations (recorded as $q_{\mathrm{productive}}, q_{\mathrm{housing}}, q_{\mathrm{consumption}}, q_{\mathrm{financial}}, q_{\mathrm{government}}$); regional variants use analogous coarse splits in their allocation tables, always normalised so that $\sum_i q_i = 1$.

Energy and entropy conventions. The internal potential $U(S_{\mathrm{M}}, V_{\mathrm{C}}, \ldots)$ is a bookkeeping scalar in currency units, and the derived quantities F_{C}, X_{C}, and G_{C} are free-energy- and exergy-like gauges constructed from $(U, S_{\mathrm{M}}, V_{\mathrm{C}})$ and fixed environmental parameters (T_0, p_0). They are interpreted as indicators of usable credit capacity and policy room, not as literal physical energies; general caveats on the analogy are collected in Section 5.

2.3 Capacity and conjugate variables

We introduce:

- V_{C}: an effective credit capacity or headroom[K4] (e.g. CET1/RWA[K3] slack, HQLA (High Quality Liquid Assets)[K6]-based lending space, liquidity buffers[K7]);
- T_{L}: a liquidity-intensity index (a "temperature-like" proxy from spreads, turnover, depth);
- p_{C}: a "credit pressure"[K5] (shadow price of capacity).

We use a quasi-potential (bookkeeping energy) with an explicit memory variable ξ that captures slow-moving structural distortions (regime, collateral and haircut conventions, concentration, funding fragility, etc.):

$$U = U(S_{\mathrm{M}}, V_{\mathrm{C}}, \xi, \ldots). \tag{3}$$

We then define reference conjugate quantities

$$T_{\mathrm{L}} \equiv \left(\frac{\partial U}{\partial S_{\mathrm{M}}}\right)_{V_{\mathrm{C}},\xi}, \qquad p_{\mathrm{C}} \equiv -\left(\frac{\partial U}{\partial V_{\mathrm{C}}}\right)_{S_{\mathrm{M}},\xi}. \tag{4}$$

Here U is not a physical internal energy; it is a bookkeeping potential chosen so that a first-law-like decomposition holds. Classical QTM/QTC do not name U; we introduce it as an analytic device. At the level of the abstract mapping, (4) should be read as a reference structure: in an idealized state description there exists some smooth potential $U_{\mathrm{th}}(S_M, V_C)$ whose canonical fields $(T_L^{\mathrm{th}}, p_C^{\mathrm{th}})$ are given by these derivatives. The empirical objects $U(t)$, $T_L(t)$, and $p_C(t)$ used in this note are proxy series constructed from GDP-like aggregates, liquidity metrics, and regulatory headroom; they are chosen to be interpretable and to track the intended directions, but they are not literally defined by finite-difference approximations to (4) in the code.

Non-equilibrium entropy balance. In an open macro-financial system, a monotonic "second law" for S_{M} is not appropriate. Instead we work with an entropy-like balance

$$\frac{\mathrm{d}S_{\mathrm{M}}}{\mathrm{d}t} = \Phi_S(t) + \sigma_S(t), \qquad \sigma_S(t) \geq 0, \tag{5}$$

where Φ_S is an entropy flow induced by external coupling (policy, foreign flows, institutional shifts) and σ_S is a non-negative entropy production rate associated with irreversible credit reallocation and market fragmentation. In this note we treat σ_S operationally and proxy it using hysteresis and loop-area diagnostics (Section 4); S_{M} itself may rise or fall depending on the sign of Φ_S.

Empirically, $U(t)$ is instantiated as a scalar credit-capacity gauge constructed from GDP-like or credit aggregates. When a dedicated U series is not available, the implementation fills it using a conservative fallback order over available GDP-like and credit-stock series (see Appendix, Internal-energy gauge). An optional detrended version $U^\star$ is used only for diagnostics and visualisation (to study fluctuations around a slowly varying baseline), while the core potentials F_{C} and X_{C} are computed from the undetrended U.

2.4 Operational metrics and irreversibility

The v2 overlay turns the thermodynamic language into directly computable operators. A minimal set of empirical ratios is

$$\eta_t \equiv \frac{\Delta Y_t^N}{C_t}, \tag{6}$$

$$d_t \equiv \frac{\Delta A_t}{C_t}, \tag{7}$$

$$\sigma_t \equiv \max\{0, 1 - \eta_t\}, \tag{8}$$

where η_t is a credit-efficiency proxy, d_t is an asset-bias proxy, and σ_t is a deliberately simple dissipation proxy. When only an asset-price index is available, a normalized alternative is

$$d_t^{\mathrm{proxy}} \equiv \frac{\Delta \log A_t}{C_t / Y_t^N}. \tag{9}$$

To give the "Thermo" label empirical bite, we also need a minimal irreversibility block. The reduced-form version used in the current v2 draft is

$$\eta_t = \eta_0 - \alpha(1 - q_t) - \beta S_{t-1} + \varepsilon_{\eta,t}, \tag{10}$$

$$S_t = \rho S_{t-1} + \gamma \frac{C_t^A}{Y_t^N} - \delta \frac{C_t^R}{Y_t^N} + \varepsilon_{S,t}, \tag{11}$$

so that asset-directed credit raises fragility while accumulated stress lowers next-period conversion efficiency.

These expressions are not presented as a complete structural law. They are a measurement-first bridge from theory to data. Existing diagnostics such as S_{M}, T_{L}, p_{C}, loop area, U, F_{C}, and X_{C} should therefore be read as companion indicators and candidate inputs into the stress block S_t, rather than as substitutes for the partition variables (C_t^R, C_t^A, q_t).

2.5 Depth and turnover proxies

In order to construct the liquidity temperature T_{L} and related gauges from observable quantities, we introduce two derived inputs: a *depth* proxy D and a *turnover* proxy Θ. They are not state variables on their own, but they are treated as diagnostic inputs whose behaviour should be consistent with the thermodynamic picture.

Conceptually, depth D captures the scale of credit outstanding relative to the system under study (e.g. private non-financial credit stocks) and therefore measures how "thick" the market is. In the absence of high-quality external depth series, we treat D as a rescaled version of real credit stocks L_{real}, so that, up to a calibration constant d_0,

$$D(t) \propto L_{\mathrm{real}}(t),$$

with the constant chosen so that the median depth over the sample is near the desired reference level.

Turnover Θ is intended to capture how quickly the system recycles capacity: it is dimensionless and increases when credit capacity is deployed and repaid more rapidly. In the simplest case we use a ratio of the form

$$\Theta(t) \propto \frac{U(t)}{L_{\mathrm{real}}(t)},$$

which compares a capacity-like quantity U with the underlying stock L_{real}. In practice this raw ratio is stabilised by replacing pathological values (division by zero, missing data) with a reference level and clipping to a reasonable band, so that extreme outliers do not dominate the temperature construction. These choices emphasise the interpretation of T_{L} as a relative index rather than an absolute physical temperature.

Design of the liquidity temperature. The liquidity temperature is designed to be high when spreads are narrow, markets are deep, and turnover is brisk, and to fall sharply when any one of these ingredients becomes stressed. To this end we combine standardized components (z_s, z_d, z_t)—the z-scores of inverse spreads, inverse depth, and turnover—into a composite

$$\hat{T}_{\mathrm{mult}}(t) = z_s(t)\, z_d(t)\, \big(1 + \tfrac{1}{2} z_t(t)\big),$$

and then rescale to $T_{\mathrm{L}} \in [0, 1]$ by min–max normalisation. In what follows we use this multiplicative construction as the baseline and denote it simply by $\hat{T}$ when no confusion arises. An additive alternative would be

$$\hat{T}_{\mathrm{add}}(t) = w_s z_s(t) + w_d z_d(t) + w_t z_t(t),$$

with weights w_s, w_d, w_t; we prefer the multiplicative form at this stage because it enforces a "weakest link" logic (a single severely stressed component can pull the product down), while still allowing turnover to modulate the combined index via the $(1 + \frac{1}{2} z_t)$ factor. The coefficient $\frac{1}{2}$ is a pragmatic choice that keeps turnover influence comparable in scale to the spread/depth product without dominating it. Appendix-style robustness checks with $\hat{T}_{\mathrm{add}}$ and other variants are natural next steps;

empirically we treat the present construction as a deliberately simple, economically interpretable starting point rather than a uniquely correct functional form, and we do not use the additive alternative in baseline indicators.

2.6 Units and scaling conventions

All core quantities are defined to have consistent, but ultimately conventional, units. The money stock M_{in} and credit stocks such as L_{real} are measured in currency units (e.g. trillions of yen); the allocation shares q_i are dimensionless, so the entropy index $H(q)$ is measured in nats and $S_{\text{M}} = kM_{\text{in}}H(q)$ has units of "currency × nats". Since $H(q)$ is $\mathcal{O}(1)$ for typical partitions, S_{M} scales essentially like M_{in}.

The capacity variable V_{C} is expressed in the same monetary units as the underlying balance-sheet aggregates (e.g. CET1-based headroom), while the pressure $p_{\text{C}} = -(\partial U/\partial V_{\text{C}})_{S_{\text{M}}}$ inherits units of "U per unit of volume". In practice U and V_{C} are both constructed from nominal stocks so that p_{C} is dimensionless and interpreted as an index of tightness; the liquidity temperature T_{L} is explicitly normalised to be dimensionless and lie in $[0, 1]$.

When regulatory headroom components are available, a representative example is the capital headroom

$$H_{\text{cap}}(t) \approx V_R(t)\,\big(1 - \alpha_{\text{cap}}\, p_R(t)\big),$$

with V_R a balance-sheet volume proxy and p_R a regulatory pressure proxy; analogous constructions apply for liquidity and funding headroom. The effective capacity is then taken as $V_{\text{C}}(t) = \min_j H_j(t)$ over such components.

The internal-energy gauge U and the derived free energies F_{C} and exergy-like X_{C} share the same nominal scale (currency units), up to the choice of the scaling constants k, T_0, and p_0. These constants are treated as knobs that set the overall magnitude of the potentials rather than objects of primary interest: in applications we focus on signs, relative levels, and changes (e.g. ΔF_{C}, X_C^+, X_C^-) rather than on any claim that the absolute values of "credit energy" carry direct economic meaning.

3 Thermodynamic Interpretation

3.1 Correspondence (QTM vs. QTC)

Table 1 summarizes the analogy. Thermodynamic quantities are literal on the left column; QTM/QTC entries are analogy-level constructs. Heat- and work-like pieces are defined for bookkeeping only.

The construction is chosen so that: (i) S_{M} is extensive and sensitive to dispersion; (ii) V_{C} carries explicit capacity/headroom information; (iii) QTC—unlike classic QTM—naturally admits a pressure-like channel p_{C}.

3.2 Conservation backbone and energy-like balance

3.3 SFC conservation backbone (accounting first law)

For early-warning use we anchor the framework in stock–flow consistent (SFC) accounting. This is the non-negotiable conservation layer: it is not an analogy but a set of identities implied by double-entry bookkeeping and flow-of-funds accounts.

Table 1: Thermodynamic–QTM–QTC correspondence (analogy-level).

Variable	Thermodynamics	QTM (money-first)	QTC (credit-first)
Mixing entropy	ΔS_{mix}	$S_{\text{M}} = kM_{\text{in}}H(q)$	$S_{\text{M}} = kM_{\text{in}}H(q)$
Temperature	T	T_L (proxy)	T_{L} (liquidity proxy)
Internal energy	U	$U_{\text{M}}(S_{\text{M}})$ (auxiliary)	$U(S_{\text{M}}, V_{\text{C}}, \ldots)$
Volume	V	–	V_{C} (capacity/headroom)
Pressure	p	–	$p_{\text{C}} = -(\partial U/\partial V_{\text{C}})_{S_{\text{M}}}$
Heat-like term	$\delta Q_{\text{rev}} = T\,\mathrm{d}S$	$Q_{\text{M}} \sim T_0 \Delta S_{\text{M}}$	$Q_{\text{C}} \sim \bar{T}_L\,\Delta S_{\text{M}}$
Work-like term	$\delta W = p\,\mathrm{d}V$	$W_{\text{M}} \equiv W_{\text{policy}}$	$W_{\text{C}} \equiv -\bar{p}_{\text{C}}\Delta V_{\text{C}} + W_{\text{policy}}$
First law	$\Delta U = T\Delta S + W$	$\Delta U_{\text{M}} = T_0 \Delta S_{\text{M}} + W + \varepsilon$	$\Delta U = \bar{T}_L \Delta S_{\text{M}} + W + \varepsilon$
Second-law tendency	$\Delta S \geq 0$ (closed, irreversible)	$\dot{S}_{\text{M}} = \Phi_S + \sigma_S,\ \sigma_S \geq 0$	$\dot{S}_{\text{M}} = \Phi_S + \sigma_S,\ \sigma_S \geq 0$
Helmholtz free energy	$F = U - TS$	$F_{\text{M}} \equiv U_{\text{M}} - T_0 S_{\text{M}}$	$F_{\text{C}} \equiv U - T_0 S_{\text{M}}$
Exergy/availability	$X \approx U - T_0 S$	$X \approx U_{\text{M}} - T_0 S_{\text{M}}$	$X_{\text{C}} = \Delta U + p_0 \Delta V_{\text{C}} - T_0 \Delta S_{\text{M}}$

Sectoral financial balances. Let $B_s(t)$ denote the net lending/borrowing (financial balance) of sector s over a period t (positive for net lenders, negative for net borrowers). Using a minimal, data-friendly three-sector aggregation $s \in \{P, G, R\}$ (domestic private, government, rest of world), SFC implies

$$B_P(t) + B_G(t) + B_R(t) = 0. \tag{12}$$

In practice B_G is available as the general-government net lending/borrowing (balance), and B_R is closely related to the current account (up to sign and conventions). When a direct private-sector net lending series is unavailable, B_P can be computed residually from (12), which is often the most robust route in public-data pipelines.

Stock–flow identity. Let $A_s(t)$ be a stock variable associated with sector s (e.g., net financial assets or a balance-sheet aggregate available at the chosen frequency). The SFC update is

$$A_s(t) = A_s(t-1) + B_s(t) + \text{VAL}_s(t), \tag{13}$$

where $\text{VAL}_s(t)$ collects valuation/reclassification effects. For monitoring, one can either (i) work with series where valuation is small at the aggregation level, (ii) treat VAL_s as an explicit residual, or (iii) use detrended / change-based diagnostics that are less sensitive to level revisions.

Interpretation inside Thermo-Credit. Equations (12)–(13) provide the hard constraint layer. The "energy-like" decomposition that follows is then a structured attribution of observed movements to dispersion (S_{M}), capacity (V_{C}), policy work, and non-equilibrium losses, while respecting the SFC accounting backbone.

When a bank grants a new loan, it simultaneously creates a matching deposit. On the joint bank–customer system, this is a pure accounting operation ("loans create deposits"), consistent with the SFC backbone in Section 3.3. Within our mapping, the associated change in the potential U is decomposed as

$$\Delta U \approx \bar{T}_L\,\Delta S_{\text{M}} - \bar{p}_{\text{C}}\,\Delta V_{\text{C}} + \bar{A}\,\Delta\xi + W_{\text{policy}} - \mathcal{D} + \varepsilon, \tag{14}$$

where:

- ΔS_{M}: change in dispersion entropy from scale and allocation shifts (typically $\Delta S_{\text{M}} > 0$ in net expansion);

Table 2: Illustrative sign patterns in expansion/contraction regimes. "Expansion with headroom use" corresponds to broad-based credit growth ($\Delta M_{\text{in}} > 0$, $\Delta S_{\text{M}} > 0$) that eats into capacity ($\Delta V_{\text{C}} < 0$). "Expansion with headroom rebuild" captures cases where asset-side growth and recapitalisation outpace use ($\Delta M_{\text{in}} > 0$, $\Delta V_{\text{C}} > 0$). "Contraction with headroom rebuild" refers to deleveraging or freezing of activity while buffers are restored ($\Delta M_{\text{in}} < 0$, $\Delta V_{\text{C}} > 0$). "Contraction with headroom use" flags stressed regimes in which activity shrinks yet capacity is still eroded ($\Delta M_{\text{in}} < 0$, $\Delta V_{\text{C}} < 0$; ΔS_{M} can be ambiguous).

Regime	ΔM_{in}	ΔS_{M}	ΔV_{C}
Expansion with headroom use	$+$	$+$	<0
Expansion with headroom rebuild	$+$	$\gtrsim 0$	>0
Contraction with headroom rebuild	<0	≤ 0	>0
Contraction with headroom use	<0	ambiguous	<0

- ΔV_{C}: change in effective capacity ($\Delta V_{\text{C}} < 0$ when headroom is used up; then $-\bar{p}_{\text{C}}\Delta V_{\text{C}} > 0$);
- $\Delta\xi$: change in a memory / distortion variable capturing slow structural shifts (regime, collateral conventions, concentration, funding fragility). $\bar{A}$ is its conjugate "affinity".
- $\mathcal{D} \geq 0$: a dissipation term capturing irreversible loss of usable credit capacity due to hysteresis, fragmentation, and forced balance-sheet adjustments.
- W_{policy}: structured policy work[K14] from regulation, guarantees, central bank operations, and other deliberate interventions;
- ε: residual term capturing model error and omitted channels.

Credit creation phases (new lending > repayments) and contraction phases (repayments > new lending) then exhibit characteristic sign patterns for $(\Delta M_{\text{in}}, \Delta S_{\text{M}}, \Delta V_{\text{C}}, -\bar{p}_{\text{C}}\Delta V_{\text{C}})$, which can be tabulated and compared with data. The value of (14) is diagnostic: it forces us to attribute changes either to dispersion, capacity use, or policy work[K14], rather than conflating them.

For intuition it is useful to summarize typical sign patterns. Table 2 collects a minimal classification in terms of net expansion/contraction and whether headroom is being used or rebuilt; the qualitative readings in the caption are meant as guides rather than axioms. These patterns provide a qualitative checklist against which observed $(\Delta M_{\text{in}}, \Delta S_{\text{M}}, \Delta V_{\text{C}})$ can be compared, and can be refined or extended in empirical work.

In addition to these pointwise diagnostics, we can study the *loop area* traced out by trajectories in the $(p_{\text{C}}, V_{\text{C}})$ plane over policy or credit cycles. In an idealized reversible limit (no dissipation, fixed ξ), the line integral would vanish for closed cycles $\mathcal{C}$, reflecting path-independence on the local equilibrium manifold. In practice, non-zero loop area measures irreversibility and dissipative effects analogous to hysteresis in magnetic systems or frictional cycles. The streaming estimator in the code updates an exponentially weighted version of this integral via

$$W_t = \lambda W_{t-1} + p_{\text{C}}(t-1)\big(V_{\text{C}}(t) - V_{\text{C}}(t-1)\big),$$

with forgetting factor λ chosen so that old cycles gradually lose influence.

Large or persistent values of W_t therefore flag regimes where policy paths in $(p_{\text{C}}, V_{\text{C}})$ space exhibit pronounced hysteresis and stress.

3.4 Dissipation and a credit quality factor Q

In the data, irreversibility shows up as non-zero loop area in the $(p_{\mathrm{C}}, V_{\mathrm{C}})$ plane. In physics, a similar idea is captured by the quality factor Q of an oscillator: it compares the energy stored in the system with the energy lost in one cycle. Here we define an analogous quality factor for the credit system.

Let loop_area(t) be the dissipative part of the policy or credit cycle in period t and let $X_{\mathrm{C}}(t)$ be the exergy-like "usable credit capacity" at time t. We first define a simple loss ratio

$$\phi(t) = \frac{\text{loop_area}(t)}{X_{\mathrm{C}}(t-1)}, \tag{15}$$

whenever $X_{\mathrm{C}}(t-1) > 0$. When $X_{\mathrm{C}}(t-1) \leq 0$, $\phi(t)$ and $Q_{\mathrm{C}}(t)$ are left undefined (treated as missing) in the baseline implementation. This tells us what fraction of available exergy is lost through policy hysteresis in that period. We then define the credit quality factor

$$Q_{\mathrm{C}}(t) = \frac{2\pi}{\phi(t)} = 2\pi\,\frac{X_{\mathrm{C}}(t-1)}{\text{loop_area}(t)}. \tag{16}$$

A large Q_{C} means that the system loses only a small share of its usable credit capacity in each cycle: policy loops move credit around with limited long-run damage. A small Q_{C} means that even modest policy loops destroy a non-trivial part of X_{C}, which is consistent with a fragile or strongly path-dependent credit structure.

For mechanical oscillators, a standard first-order approximation is $\Delta E/E \approx -2\pi/Q$. By analogy, we can write

$$\frac{\Delta X_{\mathrm{C}}(t)}{X_{\mathrm{C}}(t-1)} \approx -\frac{2\pi}{Q_{\mathrm{C}}(t)} + \text{external_injection}(t) + \varepsilon_t, \tag{17}$$

where the external-injection term collects net additions of usable credit capacity (for example, through large-scale asset purchases or recapitalisation episodes). In this way Q_{C} acts as a simple, dimensionless summary of how much hysteresis the credit system can tolerate before its usable capacity is eroded.

Interpreting W_{policy}. In the abstract decomposition (14), W_{policy} collects structured interventions (policy-rate shifts, large-scale asset purchases, regulatory changes) that are not captured by the dispersion and capacity-use terms. In the reference implementation we do not attempt to estimate $W_{\text{policy}}(t)$ separately inside the indicator pipeline; instead we compute the finite-difference terms ΔU, $\bar{T}_L \Delta S_{\mathrm{M}}$, and $-\bar{p}_{\mathrm{C}} \Delta V_{\mathrm{C}}$ and treat the residual $\Delta U - (\bar{T}_L \Delta S_{\mathrm{M}} - \bar{p}_{\mathrm{C}} \Delta V_{\mathrm{C}})$ (the "first-law residual") as a combined bucket for policy work and model error. For case studies or event analyses, users can overlay simple parametric forms—for example, specifying $W_{\text{policy}}(t) = \gamma \mathbf{1}\{t \in [t_0, t_1]\}$ around a major programme such as a large-scale asset purchase episode, or using regression-style dummies—to attribute part of the residual to known policy actions while leaving the remainder as ε.

3.5 Free Energy, Exergy, and Early-Warning Gauges

We seek a scalar gauge that (i) decreases as dispersion rises under a fixed environment, and (ii) upper-bounds structured work extractable over a cycle. A Helmholtz-style free energy provides this.

For a fixed reference environment (T_0), define

$$F_{\mathrm{C}} \equiv U - T_0 S_{\mathrm{M}}. \tag{18}$$

Then, under suitable regularity conditions,

$$\mathrm{d}F_{\mathrm{C}} = -p_{\mathrm{C}}\,\mathrm{d}V_{\mathrm{C}} + \delta W_{\text{other}}, \tag{19}$$

so that F_{C} plays the role of an available-potential measure. As $\Delta F_{\mathrm{C}} \to 0$ under chosen boundaries, policy headroom for structured adjustments is effectively exhausted.

When an ambient pressure-like parameter p_0 is also relevant, an exergy-like functional is

$$X_{\mathrm{C}} = (U - U_0) + p_0(V_{\mathrm{C}} - V_{\mathrm{C},0}) - T_0(S_{\mathrm{M}} - S_{\mathrm{M},0}), \tag{20}$$

which reduces to $-\Delta F_{\mathrm{C}}$ if V_{C} is fixed and p_0 effects are negligible. Because X_{C} depends on boundary choices (T_0, p_0), we treat it as an optional, environment-dependent gauge.

In principle X_{C} can take either sign, depending on the chosen reference state $(U_0, V_{\mathrm{C},0}, S_{\mathrm{M},0})$ and environmental parameters (T_0, p_0). Negative values indicate that the current configuration has lower "available credit energy" than the reference; for early-warning visualisation, however, it is often convenient to enforce a non-negative floor (by clipping or by subtracting the minimum value over the sample) so that zero marks the effective boundary of usable headroom.

For regime diagnostics it is also useful to separate surplus and shortage components relative to a fixed free-energy reference. Let $\Delta F_C(t) = F_C(t) - F_C^{\mathrm{ref}}$ for some chosen baseline F_C^{ref} (e.g. the first non-missing observation or a user-specified date). We then define

$$X_C^+(t) = \max\{0,\ \Delta F_C(t)\}, \qquad X_C^-(t) = \max\{0,\ -\Delta F_C(t)\},$$

so that X_C^+ tracks "excess" free energy above the reference and X_C^- tracks shortfall. These series enter plots and summary tables as complementary early-warning gauges: persistent build-up of X_C^+ suggests unusually large policy room relative to the reference regime, whereas elevated X_C^- points to sustained pressure towards the boundary of feasible adjustment.

A Gibbs-style free energy

$$G_{\mathrm{C}} \equiv U + p_0 V_{\mathrm{C}} - T_0 S_{\mathrm{M}} \tag{21}$$

can be used when (T_0, p_0) are the natural controls. In applications, we typically monitor F_{C} (fixed environment) and compare with X_{C} or G_{C} when capacity constraints are explicit (cf. exergy notions in [11]).

External coupling. In applications we also allow the credit system to be coupled to an external "pressure bath" and "temperature bath". The coupling is constructed in two stages. First, we map a small set of market driver series into standardized monthly processes. For JP, the pressure drivers are conceptually:

- the US high-yield option-adjusted spread (HY OAS), used in level form as a proxy for global credit stress;
- the US–JP 10Y yield spread, $\Delta y_{US,JP}(t) = y_{US}^{10\mathrm{Y}}(t) - y_{JP}^{10\mathrm{Y}}(t)$;
- USD/JPY log returns, $r_{\mathrm{FX}}(t) = \log \mathrm{USDJPY}(t) - \log \mathrm{USDJPY}(t - \Delta t)$.

The temperature driver is the VIX equity volatility index, again mapped to monthly frequency. Each raw series $X_k(t)$ is aggregated to month-start, transformed according to its role (level, spread, log-return), and converted to a z-score

$$Z_k(t) = \frac{X_k(t) - \mathbb{E}[X_k]}{\sigma[X_k]},$$

using sample mean and standard deviation over the available history. Second, we form composite indices by simple averaging over the available standardized drivers:

$$E_p(t) = \frac{1}{K_p} \sum_{k \in \mathcal{P}} Z_k(t), \qquad E_T(t) = \frac{1}{K_T} \sum_{k \in \mathcal{T}} Z_k(t),$$

where $\mathcal{P}$ indexes pressure drivers and $\mathcal{T}$ indexes temperature drivers. Missing values in individual Z_k are ignored on each date so long as at least one driver remains. Equal weights reflect an agnostic prior about the relative importance of each driver: with only a handful of series, more elaborate factor models or time-varying loadings would be weakly identified and risk overfitting, so the unweighted mean serves as a transparent, robust first pass. In future work one could replace this averaging with a statistically estimated factor structure or region-specific loadings once longer histories and stronger identification are available.

Given E_p and E_T we work with effective fields

$$p_{\mathrm{C}}^{\mathrm{eff}}(t) = p_{\mathrm{C}}(t) + \alpha E_p(t), \qquad T_{\mathrm{L}}^{\mathrm{eff}}(t) = T_{\mathrm{L}}(t) + \delta E_T(t),$$

so that external stress and risk-on/off conditions act as linear perturbations to the internal equation of state. When α or δ are set to zero the system is "isolated" in this sense; non-zero couplings encode how exposed the credit state is to external financial conditions.

Sectoral "chemical potentials". Given the MECE allocation shares $\{q_i\}$ and M_{in}, we can form sectoral analogues of chemical potentials. Using the normalised shares $\tilde{q}_i$ from the entropy construction, one defines

$$\mu_i = T_0 k M_{\mathrm{in}} (\log \tilde{q}_i + 1),$$

so that, up to constants, μ_i is the derivative of the entropy-like term with respect to the bucket mass. For interpretation we work with deviations from the cross-bucket mean, $\Delta\mu_i = \mu_i - \bar{\mu}$, where $\bar{\mu}$ is the average over i at each date. Large positive $\Delta\mu_i$ flag categories where incremental "credit mass" is unusually valuable in entropy terms; large negative values flag categories where allocations are already heavy relative to the rest of the system. In the current implementation these μ_i and $\Delta\mu_i$ are used as diagnostic series and are not themselves fed back into the flow dynamics.

3.6 Maxwell-Like Relation, Exterior Derivative, and Legendre Structure

In the local-equilibrium limit where $U(S_{\mathrm{M}}, V_{\mathrm{C}}, \xi)$ is approximately state-like with ξ effectively fixed over the sampling window, mixed partial derivatives commute and we obtain a Maxwell-like condition:

$$\left(\frac{\partial T_{\mathrm{L}}}{\partial V_{\mathrm{C}}}\right)_{S_{\mathrm{M}},\xi} = -\left(\frac{\partial p_{\mathrm{C}}}{\partial S_{\mathrm{M}}}\right)_{V_{\mathrm{C}},\xi}. \tag{22}$$

Outside this limit, the failure of integrability is informative rather than a fatal inconsistency.

Geometrically, the pair $(T_{\mathrm{L}}, p_{\mathrm{C}})$ defines a differential one-form on the observed $(S_{\mathrm{M}}, V_{\mathrm{C}})$ state plane,

$$\omega \equiv T_{\mathrm{L}}\, \mathrm{d}S_{\mathrm{M}} - p_{\mathrm{C}}\, \mathrm{d}V_{\mathrm{C}}. \tag{23}$$

In the integrable (state-like) regime this one-form is (approximately) exact and can be written as $\omega \approx \mathrm{d}U_{\mathrm{rev}}$ on a local-equilibrium manifold where ξ is effectively constant over the window. Outside this regime, ω should be treated as a convenient field representation rather than as the differential of a globally defined potential.

Exactness is equivalent to vanishing exterior derivative $\mathrm{d}\omega = 0$ (no "curl"), which in coordinates yields the Maxwell-like condition (22). To quantify departures from integrability, we introduce a non-equilibrium "vorticity" (curl) measure

$$\Omega(S_{\mathrm{M}}, V_{\mathrm{C}}) \equiv \left(\frac{\partial T_{\mathrm{L}}}{\partial V_{\mathrm{C}}}\right)_{S_{\mathrm{M}}} + \left(\frac{\partial p_{\mathrm{C}}}{\partial S_{\mathrm{M}}}\right)_{V_{\mathrm{C}}}. \tag{24}$$

Thus $\Omega \approx 0$ corresponds to the integrable limit (local state-like behaviour with fixed ξ), whereas persistent large $|\Omega|$ indicates strong path dependence / structural distortions and motivates modelling ξ explicitly rather than forcing a single-potential representation.

This also clarifies the Legendre structure. Starting from $U(S_{\mathrm{M}}, V_{\mathrm{C}})$ we can, in principle, define a potential written in terms of the intensive variables,

$$\Phi(T_{\mathrm{L}}, p_{\mathrm{C}}) \equiv U(S_{\mathrm{M}}, V_{\mathrm{C}}) - T_{\mathrm{L}} S_{\mathrm{M}} + p_{\mathrm{C}} V_{\mathrm{C}},$$

where $(S_{\mathrm{M}}, V_{\mathrm{C}})$ are now regarded as functions of $(T_{\mathrm{L}}, p_{\mathrm{C}})$ via the equation of state. Taking the differential and using (4) one finds

$$\mathrm{d}\Phi = -S_{\mathrm{M}}\,\mathrm{d}T_{\mathrm{L}} + V_{\mathrm{C}}\,\mathrm{d}p_{\mathrm{C}},$$

so that $(\partial\Phi/\partial T_{\mathrm{L}})_{p_{\mathrm{C}}} = -S_{\mathrm{M}}$ and $(\partial\Phi/\partial p_{\mathrm{C}})_{T_{\mathrm{L}}} = V_{\mathrm{C}}$. This Legendre transform from $(S_{\mathrm{M}}, V_{\mathrm{C}})$ to $(T_{\mathrm{L}}, p_{\mathrm{C}})$ is locally well-defined (on an approximately integrable manifold) only when (22) holds; the free-energy-type potentials introduced in §5 (such as F_{C} and G_{C}) can then be viewed as partially Legendre-transformed versions of U evaluated at fixed environmental parameters (T_0, p_0).

Empirically, one can estimate $T_{\mathrm{L}}(S_{\mathrm{M}}, V_{\mathrm{C}})$ and $p_{\mathrm{C}}(S_{\mathrm{M}}, V_{\mathrm{C}})$ from proxy series and test whether (22) approximately holds. Because U, T_L, and p_C are constructed from partially independent proxies rather than being derived from a single fitted potential, the relation is not enforced by construction; instead it functions as an integrability diagnostic. Persistent, large violations indicate non-equilibrium behaviour (path dependence / structural distortions, i.e. changing ξ) and/or proxy misspecification. In this sense, the analogy is falsifiable rather than purely rhetorical, and Section 4 summarizes a simple set of operational diagnostics built from this condition.

4 Practical Tests and Falsifiability

The mapping yields several operational diagnostics:

1. **Integrability / vorticity test.** Estimate local derivatives in (22) and compute the vorticity Ω in (24). Large, persistent $|\Omega|$ is interpreted as a non-equilibrium signal (path dependence / structural distortions), not merely "model failure".

2. **Work vs. dispersion decomposition.** Use (14) to decompose changes in U into heat-like dispersion ($\bar{T}_L \Delta S_{\mathrm{M}}$), capacity use ($-\bar{p}_{\mathrm{C}} \Delta V_{\mathrm{C}}$), and policy work ($W_{\mathrm{policy}}$).

3. **Free-energy and exergy ceilings.** Monitor F_{C} or X_{C} as early-warning gauges: sustained proximity to zero under stress scenarios flags limited room for non-disruptive adjustment.

4. **Loop area and hysteresis.** Compute loop areas in $(p_{\mathrm{C}}, V_{\mathrm{C}})$ over policy cycles. Non-zero areas capture dissipative stress and irreversibility in credit allocation dynamics[K17].

5. **Entropy production proxy.** Use loop area, $\mathcal{D}$, and changes in Q_{C} as operational proxies for the entropy production rate σ_S in (5). Rising dissipation and falling Q_{C} serve as early-warning signals of irreversible stress build-up.

All of these rely on observable or reconstructible quantities from public balance-sheet and market data. Implementation details (exact proxy choices, normalization, robustness checks) are part of ongoing empirical work and should be reported alongside results.

4.1 Testable hypotheses

The reduced-form version of Thermo Credit should be judged by a short list of predictive hypotheses rather than by conceptual elegance alone:

- **H1.** Lower q_t should predict stronger asset-price acceleration before it predicts stronger real growth.
- **H2.** Higher C_t with lower q_t should show weaker pass-through to inflation and real activity than equally large credit expansions with higher q_t.
- **H3.** High C_t^A combined with high S_t should predict greater future volatility, spread widening, or contraction risk.

This yields explicit rejection criteria. In the measurement-first reading of the model, the framework is weakened if:

- separating C_t^R and C_t^A does not improve forecasting or regime classification relative to total-credit-only baselines;
- estimated declines in q_t fail to line up with later asset acceleration or fragility build-up;
- adding the stress state S_t does not improve downside prediction beyond the existing thermo-monitoring indicators.

5 Limits, Identification, and Scope

This construction is intentionally modest. Key limitations include:

- **Category dependence.** S_{M} depends on the chosen partition of uses. Robustness across reasonable partitions must be checked.
- **Proxy noise.** T_{L}, V_{C}, p_{C} are built from noisy proxies, and U itself is instantiated from GDP- or credit-like aggregates. Measurement error or proxy misspecification can break (22) and distort the decomposition (14); in practice these relations are treated as reference structures and integrability checks rather than hard identities.
- **Local-equilibrium vs. non-equilibrium.** Fast crises and regime shifts violate smooth state-variable assumptions. We therefore interpret the Maxwell-like relations as local-equilibrium limit cases and treat vorticity and hysteresis measures as primary non-equilibrium outputs.
- **Non-physical.** We do not assume microscopic "money particles" or claim that macro-financial data obey physical first/ second laws. The mapping is analogy-level and judged only by empirical usefulness.
- **Identification challenges.** Policy, expectations, and shocks act jointly. Causal claims require careful research design beyond the bookkeeping framework presented here.
- **Partition measurement.** The coarse split (C_t^R, C_t^A) is often only partially observed in public data. In many applications, q_t must be proxy-estimated or treated as a latent state, so reported results should distinguish direct observation from inferred quantities.
- **Scaling conventions.** Constants such as k are conventional. Report normalized metrics and sensitivity checks.

This version should therefore be read as a theory-and-implementation note with explicit rejection conditions. If the partition variables (C_t^R, C_t^A, q_t) and the stress state S_t fail to improve empirical discrimination relative to simpler credit-volume baselines, the measurement-first version of Thermo Credit should be simplified or rejected in that application. A more systematic empirical evaluation for the JP configuration will follow in a later revision.

6 Data Mapping, Reproducibility, and AI Assistance

6.1 Repository-side data mapping

The measurement-first implementation layer is intentionally separated from the broader theory note. In the repository, variable meanings are fixed in `docs/definitions.md`, source and status mappings are recorded in `data/data_dictionary.csv`, and the AI-facing interface layer is outlined in `docs/thermo_credit_mcp_spec.md`. Connection patterns for local and remote AI clients are documented in `docs/mcp_connection_guide.md`, while `llms.txt` provides a short machine-facing reading order. Together these files distinguish directly observed quantities, proxy-estimated quantities, and future latent-state estimates, and they provide the shortest path from theory to recalculable outputs.

6.2 Publication stack and interface layers

The repository now separates four publication layers:

1. `report.html` for narrative interpretation and plots;
2. `docs/definitions.md` and `data/data_dictionary.csv` for fixed variable meanings and identification rules;
3. `scripts/` and `tests/` for recalculation and empirical validation;
4. an AI-facing resource/tool layer built from schemas, structured examples, and an MCP wrapper.

This fourth layer should be understood as a *distribution* mechanism rather than a validation mechanism. It makes Thermo Credit easier for AI systems to query and reuse, but it does not by itself establish the theory's empirical correctness. Validation still depends on the forecasting tests, identification strategy, and falsifiability criteria stated above.

6.3 Reproducibility and AI assistance

This note is designed to accompany an openly available codebase and data snapshots (e.g. monthly reports computed from public sources). The reference implementation lives in the repository `2025_11_Thermo_Credit` and is intended to make all reported figures and indicators reproducible.

At a high level:

- a Python pipeline computes S_{M}, T_{L}, V_{C}, F_{C}, X_{C}, and the diagnostics from public time series using shared configuration files and scripts;
- a measurement-first tool layer now exposes the compact model spec, variable definitions, core metric calculations, scenario evaluation, and regime comparisons through structured JSON interfaces;

- minimal synthetic CSVs are provided so that tests and examples can run without external API keys, while real data are fetched from standard sources (e.g. FRED, World Bank) when credentials are present;

- continuous-integration workflows (GitHub Actions) pin Python dependencies and re-run the pipeline so that indicators and the report can be regenerated from a fixed environment.

Further implementation details (module structure, configuration roles, CI layout, and synthetic data conventions) are documented in the repository README and configuration files, and can be treated as an implementation guide for practitioners wishing to reproduce or extend the system.

Drafting and editing of the present text used a GPT-based large language model as a tool at low randomness. All equations, definitions, and claims are curated and are the responsibility of the authors. Readers should rely on the archived PDF and associated repository for the citable version and reproducible code.

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Appendix: Implementation Notes (Modules & CI)

This appendix summarizes the main implementation modules for readers who wish to reproduce or extend the reference pipeline. It is not required for understanding the theoretical sections.

- **Raw and feature tables.** Region-specific builder scripts `scripts/01_build_features.py` (JP), `scripts/04_build_features_eu.py` (EU), and `scripts/05_build_features_us.py` (US) select source series according to configuration files (`config.yml`, `config_{jp,eu,us}.yml`) and `lib/series_selector.py`. They write money, credit, regulatory-pressure, and allocation tables under `data/` (e.g. `money*.csv`, `credit*.csv`, `reg_pressure*.csv`, `allocation_q*.csv`).

- **Entropy and temperature.** The module `lib/entropy.py` implements the definition $S_{\mathrm{M}} = kM_{\mathrm{in}}H(q)$ in (2) with $H(q) = -\sum_i \tilde{q}_i \log \tilde{q}_i$ where $\tilde{q}_i = \max\{q_i, 0\}/\sum_j \max\{q_j, 0\}$, and also computes $S_{\mathrm{M,in}} = kM_{\mathrm{in}}H(q)$, $S_{\mathrm{M,out}} = kM_{\mathrm{out}}H(q)$, $S_{\mathrm{M,hat}} = H(q)/\log K$ (with K buckets), and optional per-bucket contributions $S^{(i)}_{\mathrm{M,in}} = kM_{\mathrm{in}}\tilde{q}_i$. The module `lib/temperature.py` builds T_{L} from credit spreads, depth, and turnover by forming $z_s = Z(1/(\mathrm{spread} + \epsilon))$, $z_d = Z(1/(\mathrm{depth} + \epsilon))$, $z_t = Z(\mathrm{turnover})$, combining $\hat{T} = z_s z_d(1 + \frac{1}{2}z_t)$, and finally min–max normalising to obtain $T_{\mathrm{L}} = (\hat{T} - \min \hat{T})/(\max \hat{T} - \min \hat{T})$.

- **Internal-energy gauge.** Within the credit tables the column $U(t)$ is used as the internal-energy / capacity proxy entering (14), (18), and (20). When a dedicated U series is not available, the code fills it by combining available quantities in a conservative order (preferring an explicit U column, then GDP-like series such as $U_{\mathrm{gdp_only}}$ or Y, and only as a last resort falling back to L_{real}). An optional detrended version $U^{\star}$ is produced by subtracting a no-lookahead rolling mean so that diagnostics can focus on fluctuations around a slowly varying baseline.

- **Depth and turnover enrichment.** When region-specific depth/turnover series are not supplied, the helper `lib/credit_enrichment.py` constructs depth and turnover heuristically from real credit stocks L_{real} and capacity proxy U as described in the main text.

- **Measurement-first tool layer.** The module `lib/thermo_credit_tools.py` provides the compact v2 interface used by the current repository-side AI layer: theory overview, variable definitions, metric computation, scenario evaluation, and regime comparison. The script `scripts/thermo_credit_cli.py` exposes the same layer as a JSON CLI for local reproducibility checks.

- **MCP distribution wrapper.** The module `lib/thermo_credit_mcp.py` wraps the same core calculations as MCP resources, tools, and prompt templates, and `scripts/thermo_credit_mcp_server.py` provides a runnable server entrypoint. This layer is designed for AI accessibility and structured reuse; it is not a substitute for the empirical validation logic discussed in the main text.

- **Capacity, headroom, and coupling.** The core constructor `lib/indicators.py` merges money, allocation, credit, and regulatory tables, renames regulatory quantities into V_{C} and p_{C}, and, when configured with the "min_headroom" formula, sets $V_{\mathrm{C}}(t) = \min_j H_j(t)$ over headroom components. External pressure/temperature drivers are aggregated in `lib/external_coupling.py` into E_p and E_T; when the coupling coefficients α, δ are non-zero the code applies the affine shifts $p_{\mathrm{C}} \leftarrow p_{\mathrm{C}}^{\mathrm{baseline}} + \alpha E_p$ and $T_{\mathrm{L}} \leftarrow T_{\mathrm{L}}^{\mathrm{baseline}} + \delta E_T$. The same module computes F_{C}, X_{C}, and the Gibbs-style G_{C} and streams the loop-area estimator in $(p_{\mathrm{C}}, V_{\mathrm{C}})$.

- **Diagnostics.** The diagnostics routine in `lib/indicators.py` estimates $\partial S_{\mathrm{M}}/\partial V_{\mathrm{C}}|_{T_{\mathrm{L}}}$ and $\partial p_{\mathrm{C}}/\partial T_{\mathrm{L}}|_{V_{\mathrm{C}}}$ via rolling least squares, and constructs finite-difference versions of the first-law decomposition by computing ΔU, ΔS_{M}, ΔV_{C}, $\bar{T}_{\mathrm{L}}$, $\bar{p}_{\mathrm{C}}$, and the implied Q_{like} and W_{like} series.

- **Per-region orchestration.** Scripts `scripts/02_compute_indicators.py`, `scripts/02_compute_indicators_eu.py`, `scripts/02_compute_indicators_us.py`, and the helper `lib/regions.py` assemble the inputs for JP/EU/US and write `site/indicators*.csv` used by the dashboard and tests.

- **Fallbacks and CI reproducibility.** To keep tests and scheduled builds deterministic, `scripts/ci_prepare_minimal_data.py` seeds tiny synthetic CSVs when real inputs are absent, and the continuous-integration workflows (GitHub Actions) re-run the pipeline under pinned dependencies so that indicators and the report can be regenerated from a fixed environment.

Appendix: Glossary (For Physicists New to Finance)

Numbering. Each term is assigned a unique K-number (K1, K2, ...) in the order it is first referenced in the main text. Use the superscript K# markers to jump back here; headings show the same inline K#.

This glossary explains the main banking, credit, and regulatory terms used in this note for readers with a physics background. Definitions are informal and intuition-friendly. When helpful, we include analogies to thermodynamics or dynamical systems. These are aids to intuition, not strict identities.

1. Regulation and Bank Balance Sheets

K1 CET1 (Common Equity Tier 1). Core equity capital of a bank (common shares, retained earnings). It absorbs losses and underpins solvency. Analogy: the bank's fundamental "energy reserve".

K2 RWA (Risk-Weighted Assets). Total assets weighted by regulatory risk factors. Riskier exposures receive higher weights. The ratio CET1/RWA is a key prudential metric. Analogy: an effective load or mass adjusted for fragility.

K3 CET1/RWA ratio and slack. CET1/RWA indicates how much high-quality capital backs risk-weighted assets. Regulation sets a minimum. The slack is the excess above this minimum and represents capacity to expand or absorb shocks. Analogy: safety margin or unused headroom in a constrained system.

K4 Credit capacity / Headroom (V_{C}). Effective room for additional lending, given capital, liquidity, and internal limits. In this note V_{C} is the state-like variable capturing this remaining capacity. Analogy: effective volume available before hitting hard walls.

K5 Credit pressure (p_{C}). Shadow price of capacity: the marginal cost or value of relaxing or tightening V_{C}. Defined in analogy to $p_{\mathrm{C}} = -(\partial U/\partial V_{\mathrm{C}})_{S_{\mathrm{M}}}$ or $-\partial F/\partial V_{\mathrm{C}}$. Analogy: thermodynamic pressure; tighter constraints $\Rightarrow$ higher p_{C}.

K6 HQLA (High Quality Liquid Assets). Very liquid, low-risk assets (e.g. government bonds) that can be sold quickly in stress. They support liquidity ratios and crisis resilience. Analogy: high-grade stored energy that can be tapped on demand.

K7 Liquidity buffer. Cash plus HQLA held as an emergency reserve to withstand sudden outflows or market stress. Analogy: a backup battery or safety reservoir.

K8 LCR (Liquidity Coverage Ratio). Regulatory ratio: HQLA divided by projected net cash outflows over a 30-day stress scenario. It tests short-term liquidity robustness. Analogy: can the system run for 30 days under worst-case load?

K9 NSFR (Net Stable Funding Ratio). Regulatory ratio over a one-year horizon: stable funding relative to required stable funding. It limits excessive maturity mismatch. Analogy: ensuring long-lived assets are backed by sufficiently slow-decaying sources.

2. Money, Credit, and Flow Concepts

K10 Money-in-circulation. An operational measure of money that is actually circulating in the selected system and horizon (e.g. deposits used in payments), not merely base money on the central bank balance sheet. Analogy: active particles taking part in interactions.

K11 Credit stocks and flows. Stocks: outstanding amounts of loans or credit at a given time. Flows: changes per period (new lending, repayments, net issuance). Analogy: internal energy vs. power / energy flux.

K12 Margin credit / Securities financing. Credit used to finance positions in securities (e.g. margin loans, certain repos). It amplifies leverage in financial markets and is often separated from credit to the real economy.

K13 Commercial Paper (CP). Short-term unsecured debt issued by firms. Provides funding outside traditional bank loans. Analogy: an alternative branch in the credit circuit.

K14 Policy work (W_{policy}). Deliberate interventions by central banks, regulators, or governments: interest rate changes, asset purchases, guarantees, capital rule adjustments, and similar actions. In our decomposition this is the structured, intentional part of "work". Analogy: externally applied work on the system.

3. Structural and Analytical Concepts

K15 MECE (Mutually Exclusive, Collectively Exhaustive). A rule for defining categories: (i) no overlaps between categories, (ii) no gaps in coverage. For the entropy $S_{\text{M}} = kM_{\text{in}}H(q)$, the allocation shares q_i should be MECE, otherwise S_{M} is distorted by double counting or omissions.

K16 State variable and proxy. A state variable is determined by the current state, not by the detailed path. In practice, finance uses observable proxies (e.g. ratios, indices) to approximate such variables. Our Maxwell-like relation (22) tests whether chosen proxies behave as if they came from a consistent potential $U(S_{\text{M}}, V_{\text{C}}, \ldots)$.

K17 Hysteresis and loop area. If trajectories in the $(p_{\text{C}}, V_{\text{C}})$ plane form loops with non-zero area over policy or credit cycles, this indicates irreversibility or dissipation-like effects (e.g. stress, misallocation, or path dependence). Analogy: magnetic hysteresis, frictional cycles, or inelastic processes.

K18 Stress testing and early-warning indicators. Stress tests simulate severe but plausible scenarios to see whether banks or systems survive. Early-warning indicators attempt to flag fragility in advance. Within this framework, quantities like F_{C}, X_{C}, and loop areas are intended as structured candidates to complement such tools.

K19 Money vs. credit (QTM vs. QTC perspective). In traditional Quantity Theory of Money (QTM), money is primary and credit is often treated as derived. In the Quantity Theory of Credit (QTC) and in this note, credit creation is primary: bank lending creates deposits, and money is an accounting outcome of credit decisions. This shift motivates tracking credit capacity and dispersion as state-like objects.

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